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MODEL SELECTION AND VALIDATION REPORT ( general_biomass model )
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## OBSERVATIONS:3451

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( general_biomass model )

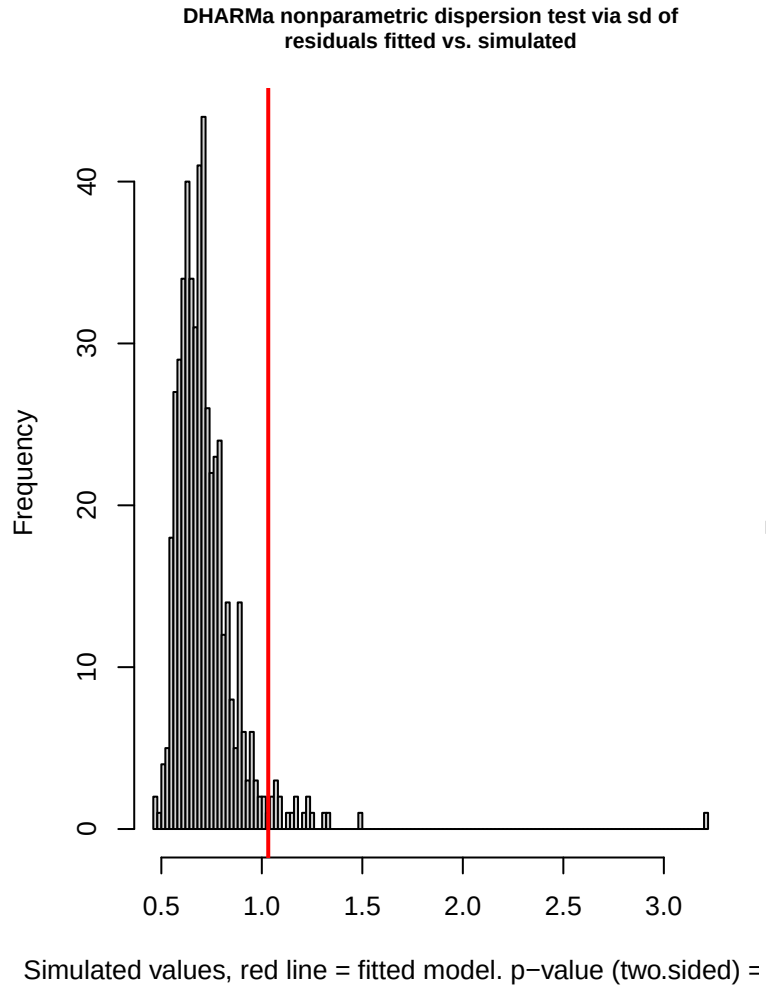
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

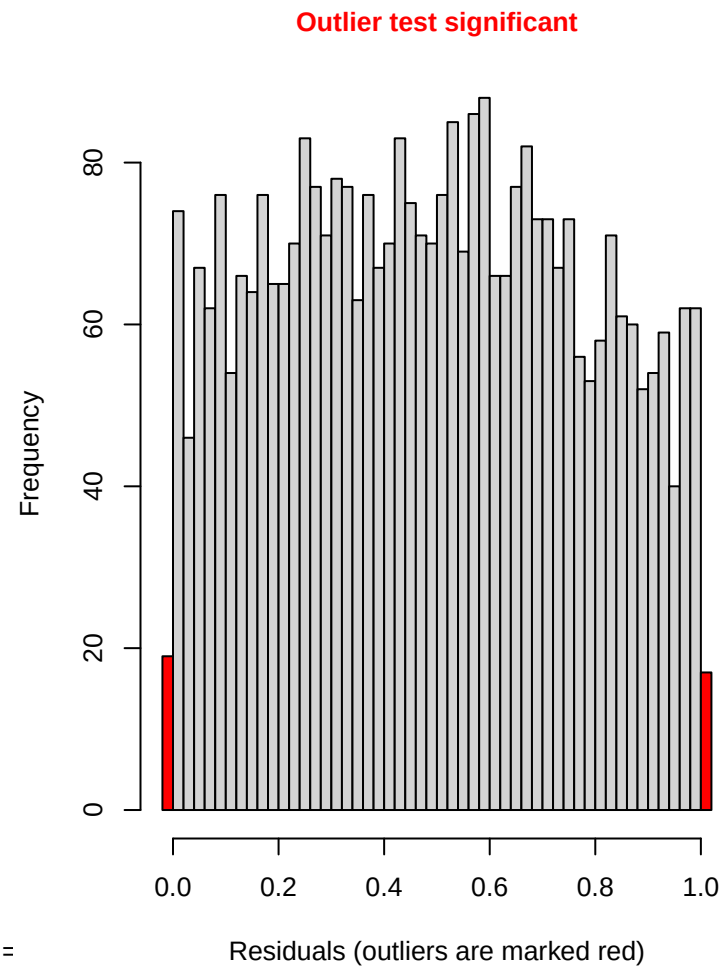
Conditional model:
              coef.est coef.se
(Intercept)       7.03    0.19
kde_fric          0.07    0.02
kde_feve         −0.17    0.02
tric              0.05    0.02
teve             −0.32    0.02
sbt_mean          0.33    0.06
sic_mean          0.03    0.05
sbt_sd            0.11    0.03
sst_sd            0.04    0.03
log_chla_mean     0.03    0.02
depth            −0.01    0.03

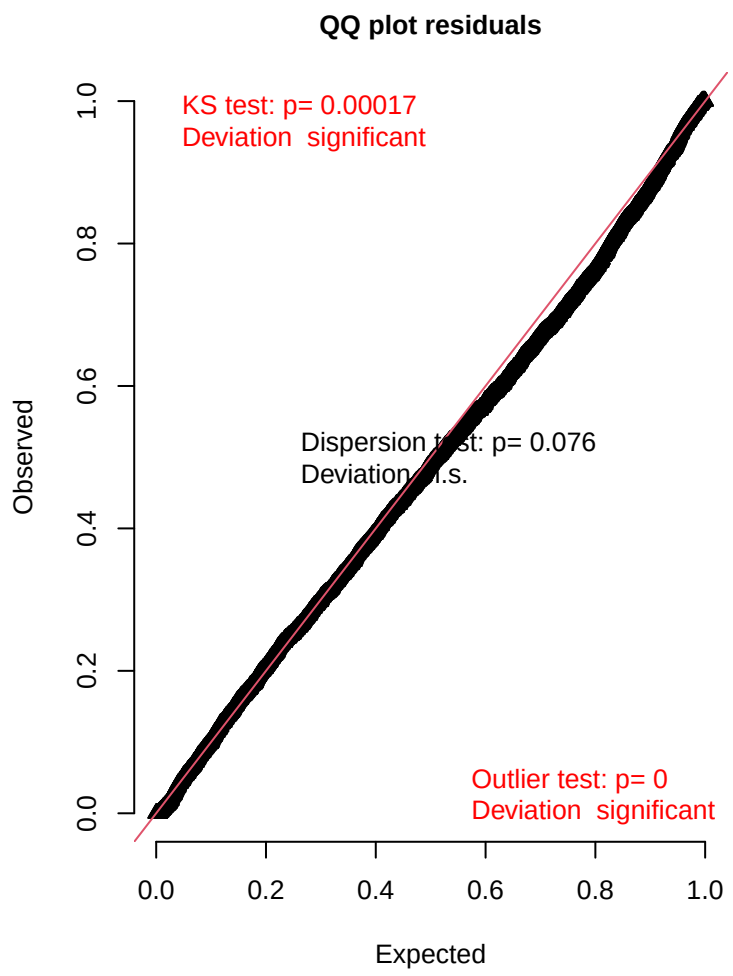
Dispersion parameter: 0.76
Spatiotemporal AR1 correlation (rho): 0.88
Matérn range: 296.83
Spatiotemporal marginal AR1 SD: 0.86
REML criterion at convergence: 27625.194

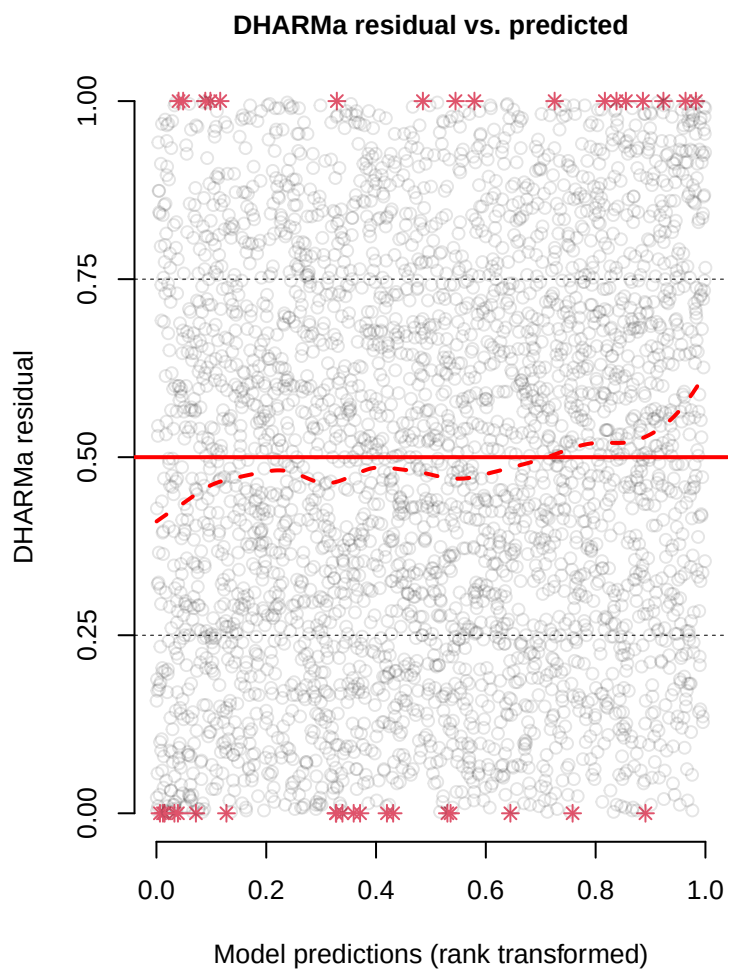
See ?tidy.sdmTMB to extract these values as a data frame.

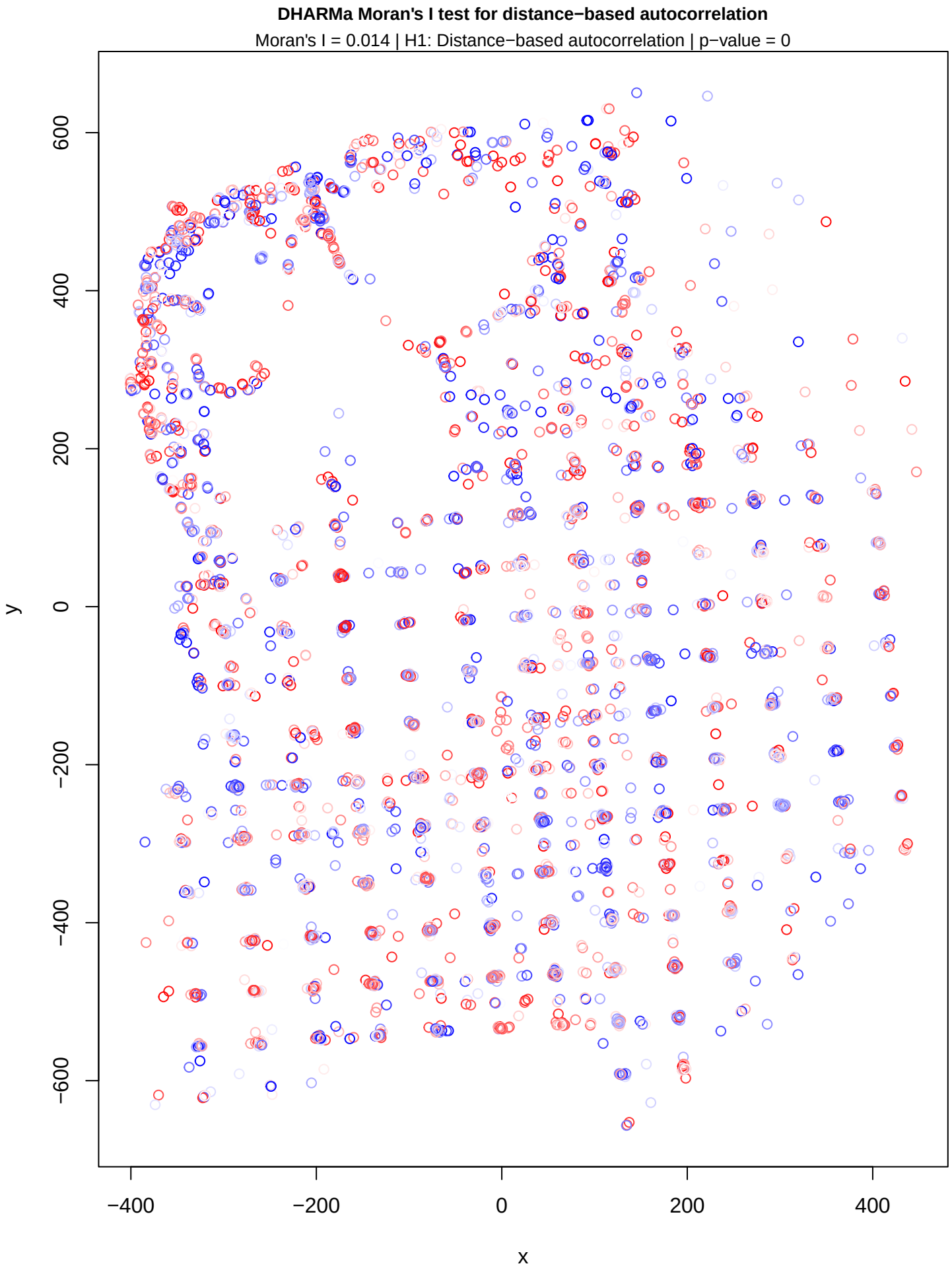
## SELECTION STATISTICS
                                                   df      AIC   delta_AIC
spatiotemporal_random_fields_ar1                   15 55280.39   0.4243022
spatiotemporal_random_fields_ar1_with_anisotropy 17 55280.81 295.1350364
spatiotemporal_random_fields_iid_with_anisotropy 16 55575.95   3.1753635
spatiotemporal_random_fields_iid                   14 55579.12 302.0616787
spatial_random_fields                              14 55881.18   3.7466393
spatial_random_fields_with_anisotropy              16 55884.93 655.6846007
error_only                                         12 56540.62          NA
```

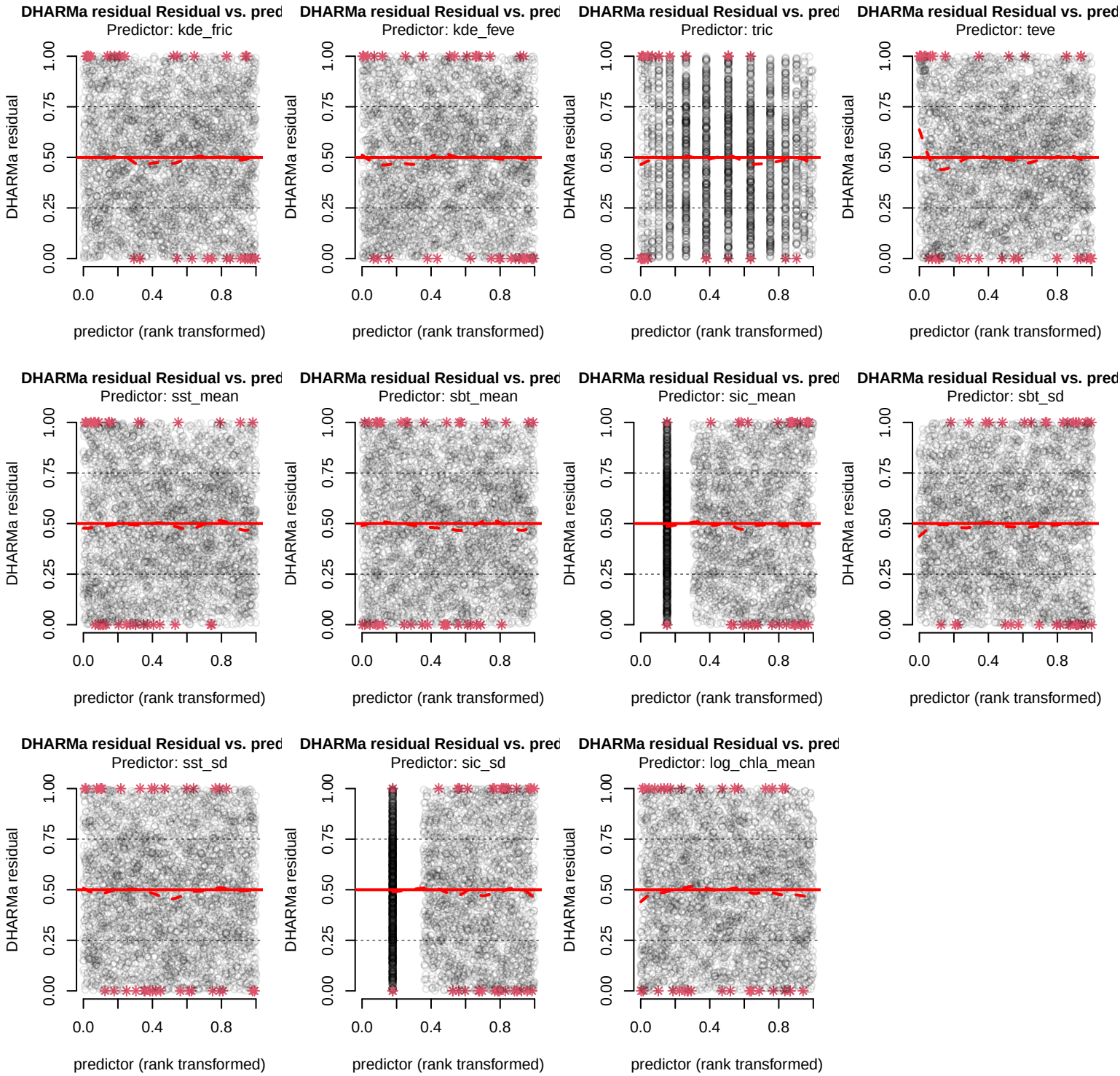

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
10
20
30
40
0.5
1.0
1.5
2.0
2.5
3.0
Simulated values, red line = fitted model. p−value (two.sided) =


Outlier test significant
Frequency
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)


QQ plot residuals
KS test: p= 0.00017
Deviation significant
Dispersion test: p= 0.076
Deviation n.s.
Outlier test: p= 0
Deviation significant
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected


DHARMa residual vs. predicted
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.014 | H1: Distance−based autocorrelation | p−value = 0
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: kde_fric
Predictor: kde_feve
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8